MAGNETIZATION OF THE COLUMBIA RIVER BASALTS IN WASHINGTON AND NORTHERN OREGON

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(Received May 7, 1956)

ABSTRACT

The remanent magnetization of late-Tertiary lavas in seven sections of Columbia River basalts has been examined. Approximately equal numbers of normally and reversely magnetized lava flows have been found. At least three field reversals during their eruption are indicated, if the reversed magnetization is interpreted as the result of a reversed geomagnetic field at the time of cooling of the lavas.

Introduction

The sampling and testing of this preliminary study were undertaken for three reasons: first, in order to throw light upon the causes of anomalous magnetization in lavas; second, to investigate the possibility of using the remanent magnetization of the flows as a basis for stratigraphic correlation; and, third, to provide suites of oriented sections for more general systematic petrologic studies of the basalts in the future.

The Collection and Preparation of the Specimens

In seven areas of nearly horizontal basalts, some 114 flows aggregating 11,350 feet in thickness were examined. Oriented fist-sized samples were collected from those flows in which the remanent magnetism seemed likely to be least disturbed by chemical alteration or by movements after hardening. The choice of sampling areas was intended to give evenness of geographical distribution in the basalt series and, in the group of three eastern areas, comparisons within single wide-ranging flows. The number of samples that were collected from each flow, however, are hardly sufficient to investigate the effects of lightning strikes on the outcrop, the effects of reheating of flow tops by the later flows, or any contrasts between rinds and centers of flows. These matters need careful study in the future.

The present collecting was done with the aid of a Brunton pocket transit and cannot be guaranteed to be more accurate than about three degrees. Preliminary tests made next to cliffs of various types of basalt indicated declination changes of perhaps two degrees at a distance of three feet. The specimens were easily-detached joint blocks of suitable size and freshness, marked with "dip and strike" lines

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penciled on adhesive strips and measured before being lifted away from the outcrop. Most of these specimens deflected the magnetic needle less than one degree when held next to it.

In the laboratory, vertical cores were drilled from the specimens. These were then sliced to make cylinders, approximately $1\frac{1}{4}$ inch in diameter and $1\frac{1}{4}$ inch long. The orientation marks were transferred to the horizontal surface.

The always debatable distinction between flows and flow-units was borne in mind during the collecting but was not an important one for this preliminary study. More probably two or more true flows were occasionally taken to be one single flow in places where crumbling vesicular tops were concealed by soil and grass on flatter slopes.

Measurement of Direction of Magnetization

The angles of declination and dip of the magnetic polarization of the cylindrical samples were measured on a simple astatic magnetometer. Such an instrument, suitable for the measurement of igneous rocks, has been described by Collinson, Creer, Irving, and Runcorn [see 1 of "References" at end of paper]. An accuracy of 1° is aimed at, but the errors in collection are probably about 2° to 3°, and the scatter of directions within a single flow is often 5° to 10°, but may be much larger occasionally, for reasons not at present understood.

TABLE 1—*Directions of magnetization of lava flows*

Series of flows	Normally magnetized				Reversely magnetized			
	D	*I*	α	κ	*D*	*I*	α	κ
		°	°			°	°	
Imnaha River (Wallowa Co., Oregon)	N. 30.2°E.	+70.9	14.1	5.0	S. 9.0°E.	−69.4	21.6	3.4
Grande Ronde River (Asotin Co., Wash.)	N. 39.0°W.	+85.4	24.1	3.1	S. 51.1°E.	−71.7	28.2	2.9
Steptoe Canyon (Whitman Co., Wash.)	.....	.....	.....	.....	.....	.....	.....	.....
Blue Lake (Grant Co., Wash.)	N. 6.0°W.	+70.4	14.8	7.0	S. 19.0°W.	−67.2	17.6	7.7
Wallula Gap (Umatilla Co., Ore.)	N. 13.0°E.	+58.1	8.7	26.2	.....	.....	.....	.....
Union Gap (Yakima Co., Wash.)	N. 0.2°E.	+67.7	30.0	3.5	S.20.5°W.	−56.9	.....	.....
Multnomah Falls (30 mi. E. of Portland)	N. 3.9°E.	+39.4	.....	.....	S.15.7°W.	−47.6	18.9	17.4
Mean of all flows	N. 11.4°E.	73.5	7.6	4.6	S. 3.1°E.	−66.0	10.7	4.2
Present direction of geomagnetic field	N. 20° to 22°E.	+70° to 72°						
Dipole direction	N. 0°	69°						

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As is found for other Tertiary lava flows (see [2], for example), the directions of magnetization can be divided into two groups—one normal, scattered about the direction of the present earth's field, and one reversed, antiparallel to the other group. Table 1 shows the mean directions of magnetization of the normal and reversed flows from the seven different sites at which the lava series were sampled. The statistical treatment is that due to Fisher [3]; α is the semi-angle of the cone described about the mean direction which includes to a probability of 95 per cent the true direction, while κ is a measure of the precision. Stereograms 2 to 7 show the directions of magnetization of each sample measured. The magnetization of a single sample from a flow will usually diverge considerably from the mean direction of magnetization of a flow. This study, therefore, throws no light on the secular variation of the geomagnetic field such as would be obtained if the mean direction of magnetization of the separate flows were known. In the sections shown in Figure 1, directions are classified as normal (N) or reversed (R), or are left

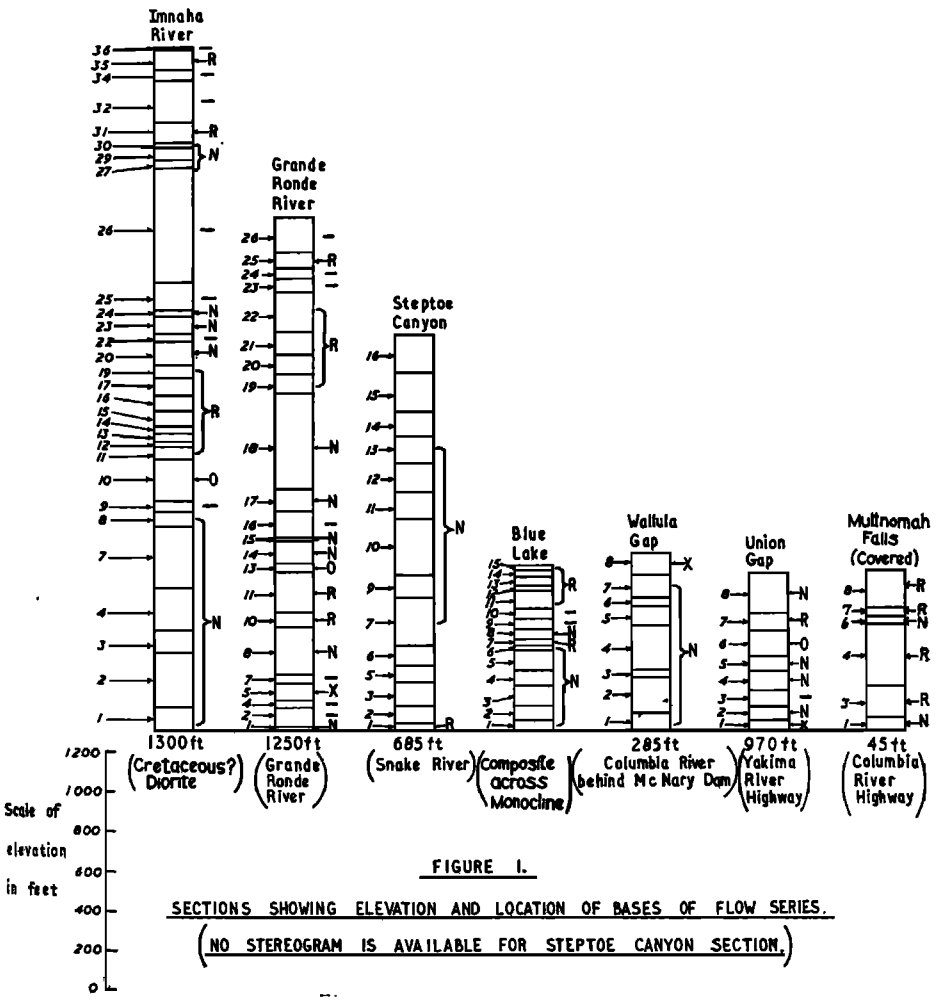
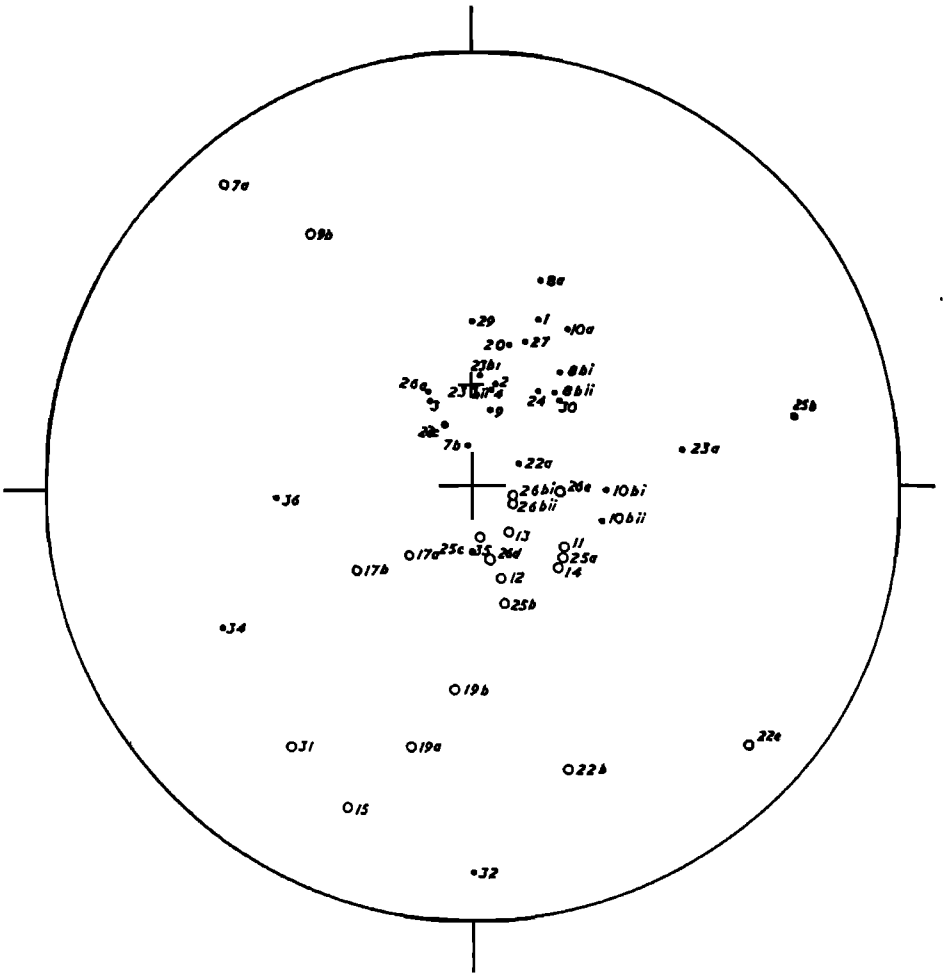


FIGURE 1.

SECTIONS SHOWING ELEVATION AND LOCATION OF BASES OF FLOW SERIES.

(NO STEREOGRAM IS AVAILABLE FOR STEPTOE CANYON SECTION.)

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Imnaha River Basalt Series, Wallowa County, Oregon.

FIGURE 2.

○ denotes north pole on upper hemisphere of stereographic projection.
• denotes north pole on lower hemisphere of the projection.

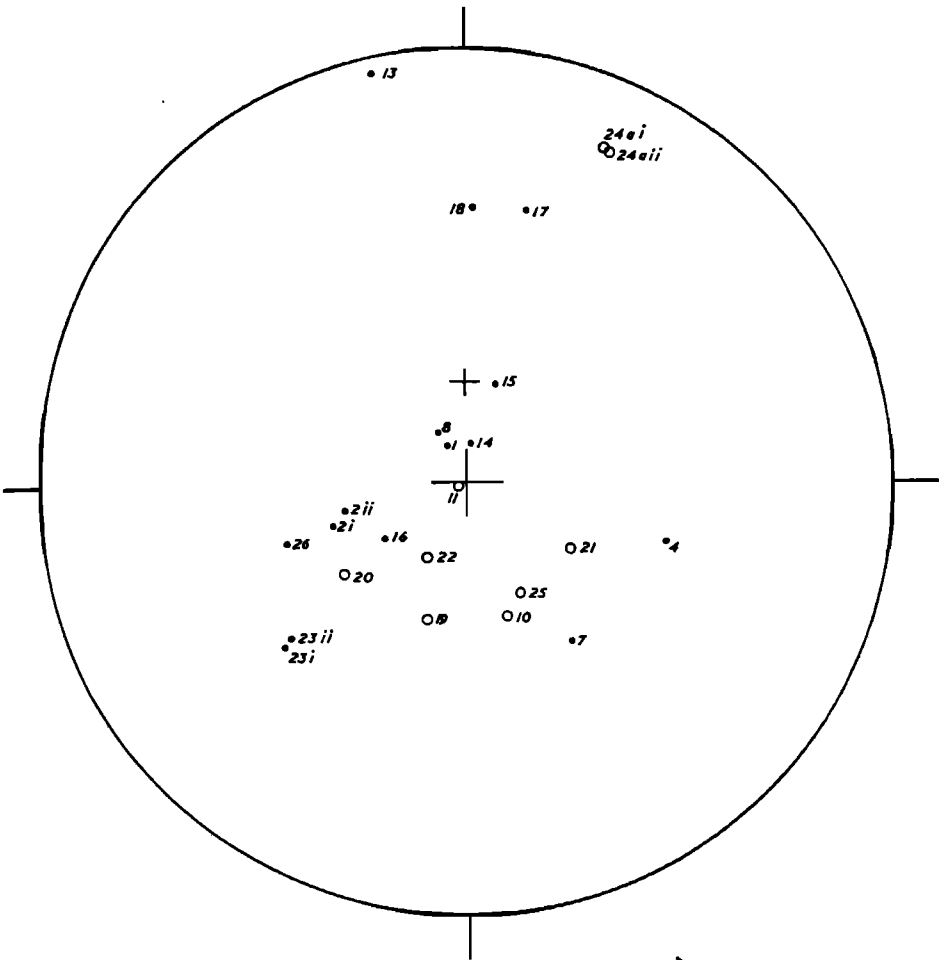
unclassified (—) if these are not in reasonable accord with the predominant directions. *X* denotes flows not sampled. Lavas are numbered from the base upwards. Letters after the lava number denote separate samples from the same flow; numerals denote separate cores from the lava sample.

The directions of magnetization of the normal and reversed group are not significantly different from those of a geomagnetic dipole field with the polarity of the present field and its opposite, respectively.

Microscopic Examination of Specimens

The 138 samples collected for magnetic testing were also examined petrographically and in reflected light, in order to discover any correlations existing

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Grande Ronde River (Rattlesnake Creek) Basalt Series, Asotin County, Washington.

FIGURE 3.

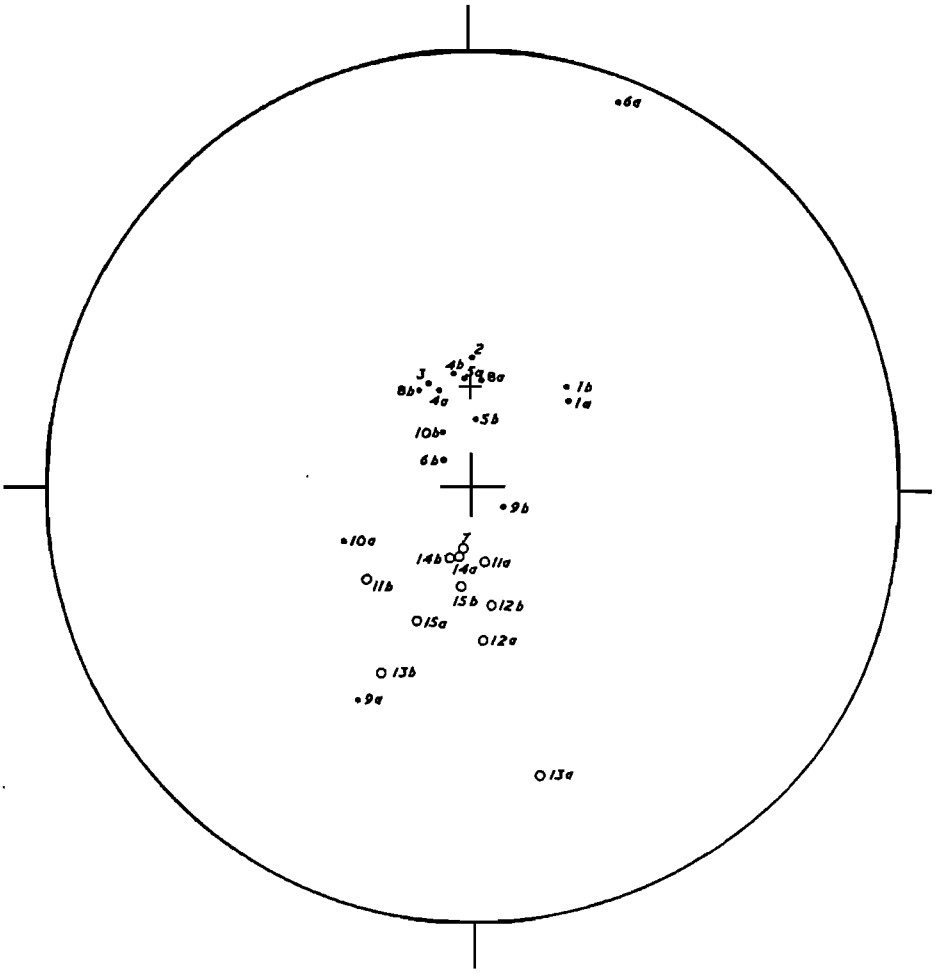
between content of ilmenite and the magnetic polarity of the rocks, such as has been suggested by the work of Balsley and Buddington in the Adirondacks [4]. No chemical analyses were made which would show the total non-silicate titanium, of which much might be present dissolved in the abundant magnetite of the more rapidly cooled lavas. The results of the microscopic work are summarized by histograms shown in Figures 8 and 9.

The following general relationships were noted:

(1) The content of magnetite is usually greatest (10 to 20 per cent) in rocks with the greatest per cent (40 to 80) of glassy residuum or its devitrified equivalent; in the glass, magnetite assumes dendritic forms, and in the dense devitrified rocks it forms a peppering of unjoined minute crystals.

(2) Ilmenite is more abundant in the basalts with a low content (under 10

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Blue Lake Basalt Series, Grant County, Washington.

FIGURE 4.

per cent) of glassy residuum and therefore a low content of magnetite (under 5 per cent). The maximum ilmenite seen was 5 per cent.

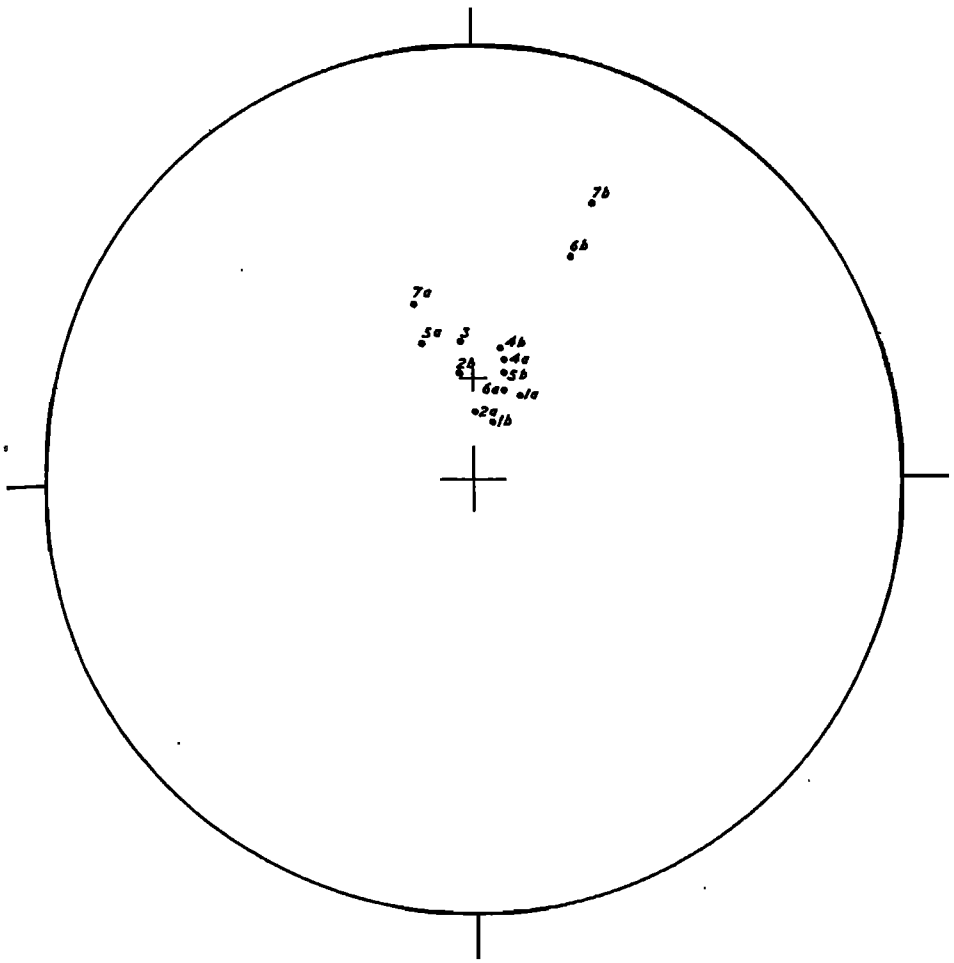
(3) There is no clear relationship between the percentage of glass and average diameter of crystals other than phenocrysts.

(4) Of all specimens tested magnetically, 65 per cent are normally polarized; of the ilmenite-bearing ones only, about 60 cent are normally polarized; of those containing 30 per cent or more of glass—that is, those presumed to have chilled most rapidly—about 45 per cent are normally polarized.

Conclusion

The number of normally magnetized flows found was 44, and the reversely

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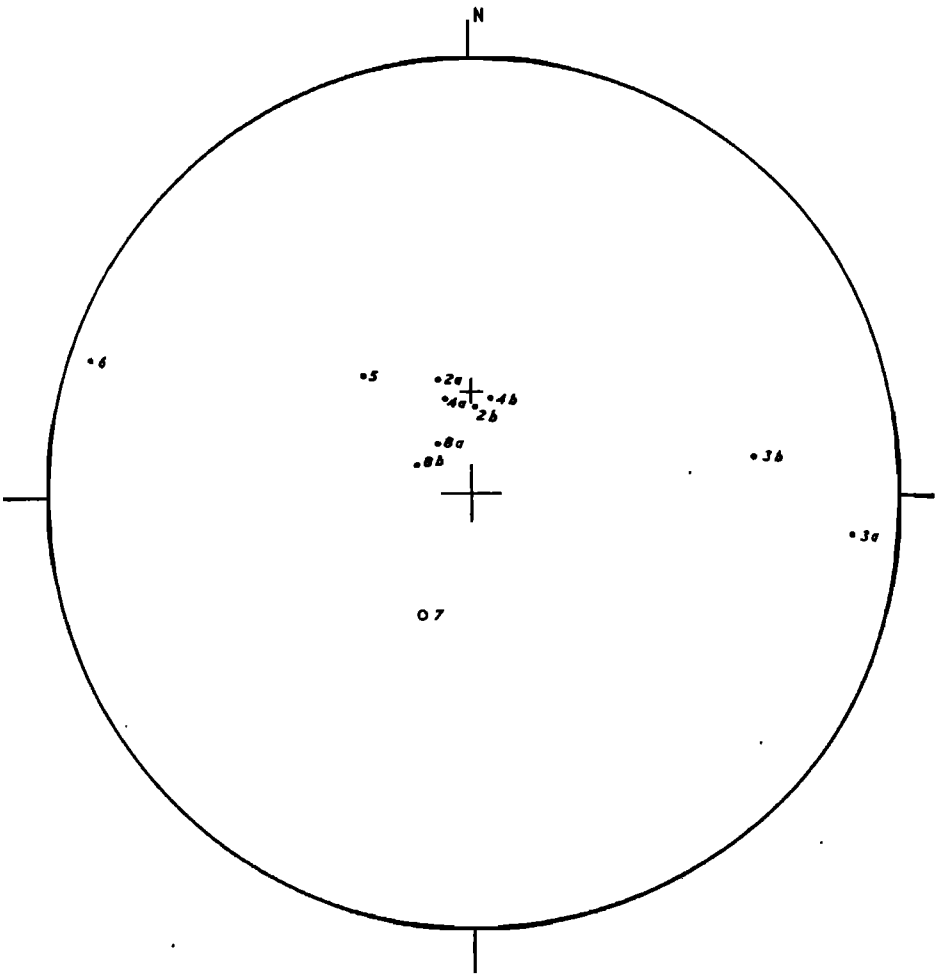
Wallula Gap Basalt Series, Umatilla County, Oregon.

FIGURE 5.

magnetized flows numbered 29—a similar ratio to that observed in Iceland. The mean number of flows between successive reversals is 3.9. This is considerably less than the figure for the Icelandic lavas and possibly indicates a much slower mean rate of extrusion of flows, offsetting a greater mean thickness of 98 feet per flow.

These basaltic flows are of Miocene and perhaps Pliocene age [5]. If the reversed magnetization is interpreted as having arisen from a field opposite to the present one, the evidence from the Grande Ronde and Multnomah Falls series indicates that at least three field reversals occurred during the time of eruption. It has been urged that reversed magnetization is due to anomalous properties of the iron-titanium oxide minerals and not to reversals of the geomagnetic field. The petrological examination reveals no obvious difference between reversed and normal flows. If this is the true explanation, anomalous properties must be present to an

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Union Gap Basalt Series, Yakima County, Washington.

FIGURE 6.

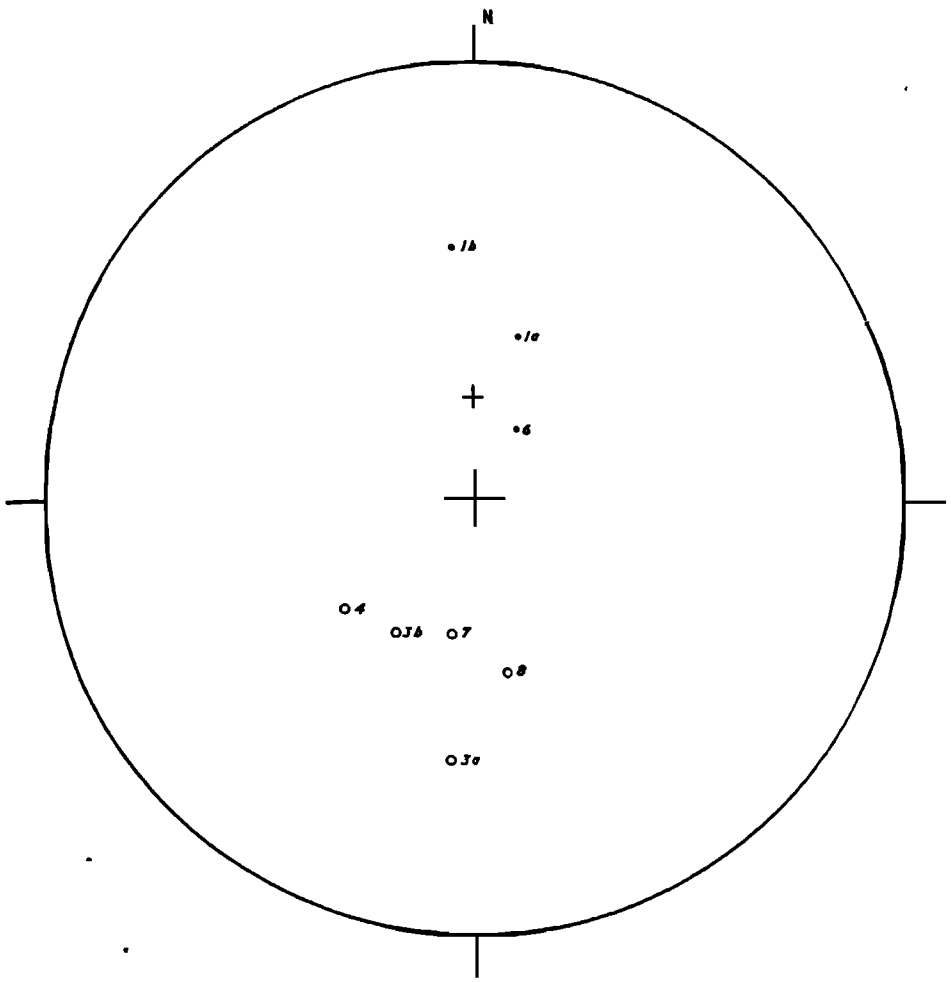
approximately equal degree in the Columbia River basalts and the Icelandic basalts.

Flow 10 of the Imnaha series appears to have a magnetization markedly different from either the well-grouped series of six normal flows below or the seven reversed flows above. Flow 6 in the Union Gap seems similarly oblique to the prevalent directions. These flows may therefore have been extruded during the time in which the field was changing between its two stable polarities. Cases of such intermediate magnetization need further study.

Acknowledgments

The initiation of this work and preliminary measurements of the specimens was done by one of us (S.K.R.) during the summer of 1953 while he was a research

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Multnomah Falls Basalt Section, 30 miles east of Portland, Oregon.

FIGURE 7.

geophysicist at the Institute of Geophysics of the University of California at Los Angeles; he is grateful to Drs. A. C. Waters and R. E. Wilcox for an initiation into the geology of the Columbia River lavas. We are much indebted to the Director of the Institute, Dr. L. B. Slichter, for his interest. Funds for the collection, shipping, and sectioning of samples were provided by the State College of Washington, as Project 85 of the College Committee on Research. We are indebted for field assistance to Mr. and Mrs. A. Keppler and Messrs. F. Howd, A. Parmer, Q. Aune, H. Praetorius, and R. Page; to Mr. Howd for work on polished sections; and to Mr. G. Turnbull for help with the measurements and computations.

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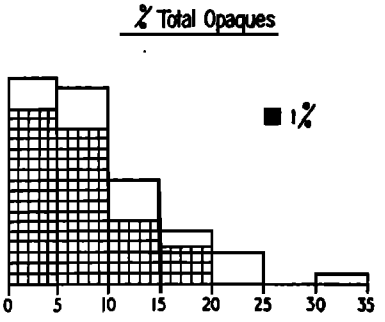


FIGURE 8.

Normally Magnetized Flows.

Reversely Magnetized Flows.

Histogram of percentage of total opaques

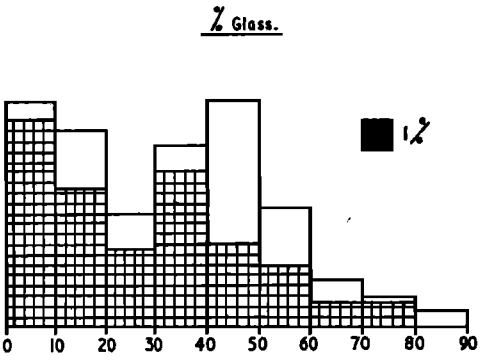


FIGURE 9.

Normally Magnetized Flows.

Reversely Magnetized Flows.

Histogram of percentage of glass.

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